EMS619U-BIOMEDICAL DEVICE DEVELOPMENT

A CRITICAL EVALUATION OF THE STATE OF THE ART IN FRACTURE FIXATION

Name: Yemesis Addo
Program: BEng Biomedical Engineering
Student number: 220512826

EXECUTIVE SUMMARY

Fracture healing follows either primary or secondary healing, and is influenced by several physiological, pathological, and anatomical factors, with emphasis on the need for osteogenic cells, growth factors, scaffold materials, and mechanical stability in successful bone regeneration. This report reviews the current state of the art of fracture healing including current treatment techniques and biomaterials being used in fracture fixation, patient-specific factors that may affect fracture healing outcomes, issues resulting in impaired bone healing and advances in relevant research and technology.

Fracture repair methods are usually effective in restoring bone back its to pre-traumatic cellular and mechanical state, however, 10-15% of all fractures fail to heal due to issues such as non-unions (occurring in 2-10% of all fractures and up to 50% in tibial fractures), infections (up to 30% in open fractures), cortical porosis, implant failure, and refractures (Up to 5% of fractures).

Traditional surgical repair methods such as internal and external fixation and bone grafting remain dominant in fracture healing, however, post-operative issues including stress shielding, non-unions and malalignment remain a challenge. Conservative and pharmalogical therapies such as LIPUS and ESWT, are other fracture repair methods currently being explored, however issues with varying effectiveness remain a concern, and these methods lack sufficient quality human trials, making it hard to validate their efficacy. Titanium and stainless steel are dominant in use for fracture fixation but issues with cost and toxicity remain a challenge with such metals. Polymers and other bioabsorbable materials are also being used however challenges with strength and unpredictability remain an issue in their usage in fracture fixation.

Experimental tests show that fatigue failure often occurs at the screw-plate interface, pointing to the need for better fixation designs. Despite advances in mechanical testing, there is a need to establish standardised tribological tests to assess debris generation to help lower the risks of osteolysis, inflammation, and aseptic loosening. Future studies should include long-term clinical assessments and more biocompatible materials to improve patient recovery in fracture healing.

THE BIOLOGY OF FRACTURE HEALING

A fracture is a complete or incomplete break in the anatomic continuity of bone, leading to mechanical instability and often limited mobility (Bigham-Sadegh and Oryan, 2014). Fractures may be classified on many bases but are generally described as closed (no skin break) or open (skin break). Open fractures are associated with more damage to surrounding soft tissues and higher risk of infection. Long-bone fractures can be classified according to the forces from which they result. Simple fractures result from twisting or bending forces, resulting in two bone fragments, and comminuted fractures result from high-velocity injuries, resulting in several fragments. (Oryan et al., 2014).

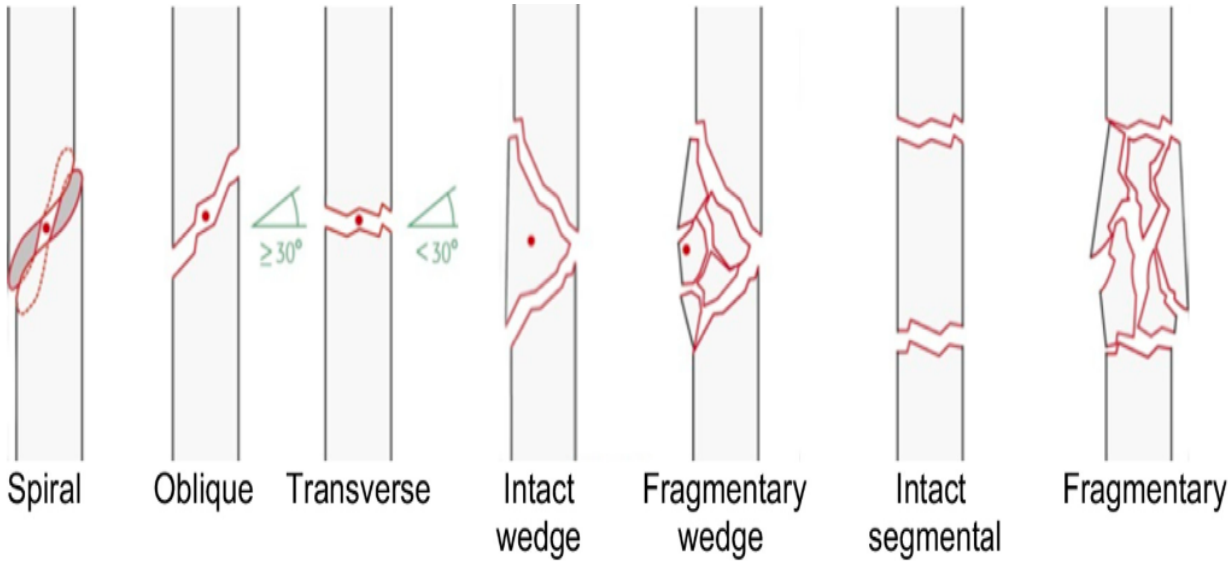


Figure 1: *Types of fractures (Parra-Cabrera, Pérez-Cano and Jiménez-Delgado, 2022).*

Fracture healing is the process of bone regeneration post-trauma (Marsell and Einhorn, 2011). According to P.V. Giannoudis, the Diamond Concept comprises four basic factors necessary for successful bone regeneration: a healthy population of osteogenic cells, growth factors, an osteoconductive scaffold, and a mechanically stable environment. Other factors like age, medical history, and underlying conditions like osteoporosis could also influence bone regeneration. There are two possible bone healing pathways post-trauma: Primary/Direct healing, which requires rigid stabilisation and occurs without callus formation, and Secondary/Indirect healing (*Figure 2*) which is the more common pathway and the natural bone healing process (Marsell and Einhorn, 2011). It comprises inflammation, repair and remodelling phases summarised in *Table 1* (Kim et al., 2020). During remodelling, bone adapts to mechanical stress according to Wolff's Law, reinforcing areas under higher loads while resorbing bone in low-stress regions (Frost, 1987).

Table 1: *Phases of Fracture Healing (Kim et al., 2020).*

Phase	Key Processes	Cellular Activity	Outcome
Inflammation	• Hematoma formation • Recruitment of inflammatory cells • Vascularisation	Macrophages, neutrophils and osteoprogenitor cells infiltrate the fracture site	Formation of a vascular network and initiation of healing
Repair	• Formation of a fibrocartilaginous soft callus • Hard callus mineralisation	Chondrocytes, fibroblasts, and osteoblasts deposit new bone through endochondral and intramembranous ossification	Increased mechanical stability and restoration of bone continuity
Remodelling	• Maturation of woven bone • Refinement of bone structure	Osteoclasts resorb excess bone, while osteoblasts refine the trabecular architecture	Full restoration of bone integrity and mechanical function

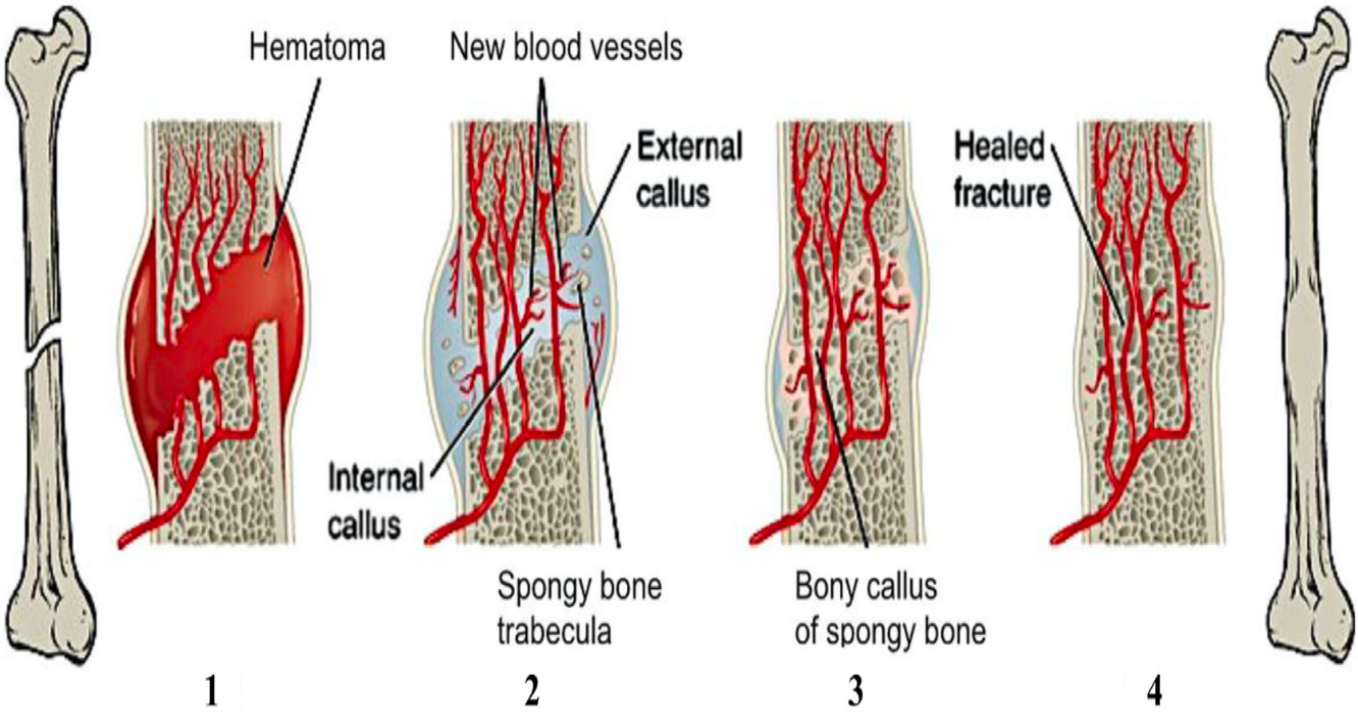


Figure 2: *Natural Healing Process of Bone:* ***(1)*** *hematoma formation,* ***(2)*** *soft callus formation,* ***(3)*** *hard callus formation, and* ***(4)*** *remodelling (Kim et al., 2020).*

CURRENT STATE OF THE ART IN FRACTURE REPAIR METHODS

Fracture Fixation (A Brief History)

Fracture fixation is the process of stabilising fractured bone to facilitate healing and restore mobility. There are two main types of fracture fixation: internal and external (Compared in *Table 2*). Internal fixation uses surgical procedures and devices to stabilise fractured bone internally, while external fixation uses rigid frames screwed into fractured bone. (Taljanovic et al., 2003). Bone implants can fail by breakage or fatigue if they take over mechanical load from the peri-implant bone (stress shielding), leading to bone resorption due to insufficient stimulation needed for bone remodelling (Kim et al., 2020).

Early plate designs were abandoned due to challenges with corrosion and insufficient strength. Danis (1949) pioneered compression plating and influenced all subsequent plate designs. Limited Contact-Dynamic Compression Plates (LC-DCP) and Point Contact Fixators (PCF) were designed to reduce cortical porosis caused by early designs of Dynamic Compression Plates (DCP), by reducing bone-plate contact. Clinical trials by Field et al. (1997) and Jain et al. (1999) however challenged claims of reduced contact area and concluded that neither innovation was advantageous in fracture healing over the DCP (Uhthoff, Poitras and Backman, 2006).

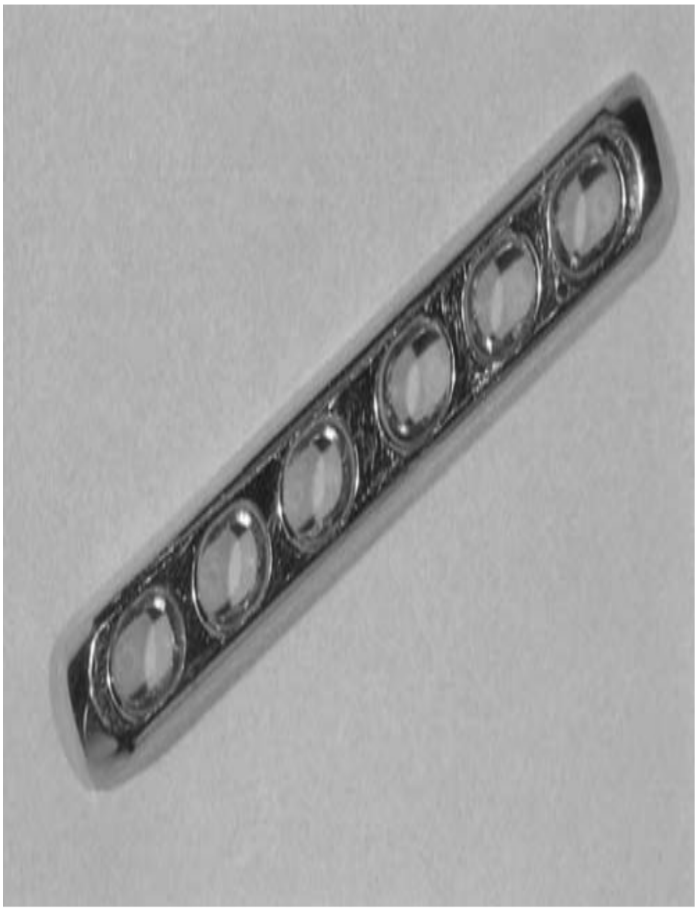


Figure 3: *Dynamic Compression Plate (DCP). (Uhthoff, Poitras and Backman, 2006).*



Figure 4: *Radiograph of the wrist showing a Low Contact Dynamic Compression Plate (LC-DCP) (Taljanovic et al., 2003).*

Table 2: *Comparison of Internal and External Fracture Fixation (Hannigan et al., 2015).*

Type of fixation	Internal Fixation	External Fixation
Types	• Plates and Screws • Intramedullary (IM) Nails/Rods • Wires and Pins	• Standard uniplanar fixators • Ring fixators • Hybrid fixators
Advantages	• Provides rigid stability for fracture healing • Reduces risk of malalignment • Prevents excessive motion at fracture site • Better for upper extremities	• Allows rapid stabilisation • Non-invasive • Minimises damage to soft tissue • Better for temporary fixation
Disadvantages	• Invasive • May cause stress shielding • Periosteal blood supply disruption	• Higher risk of pin-track infections • Limited rigidity • May cause malalignment



Figure 5: *Frontal (left) and lateral (right) radiographs of the knee showing a tension band wire (Taljanovic et al., 2003).*

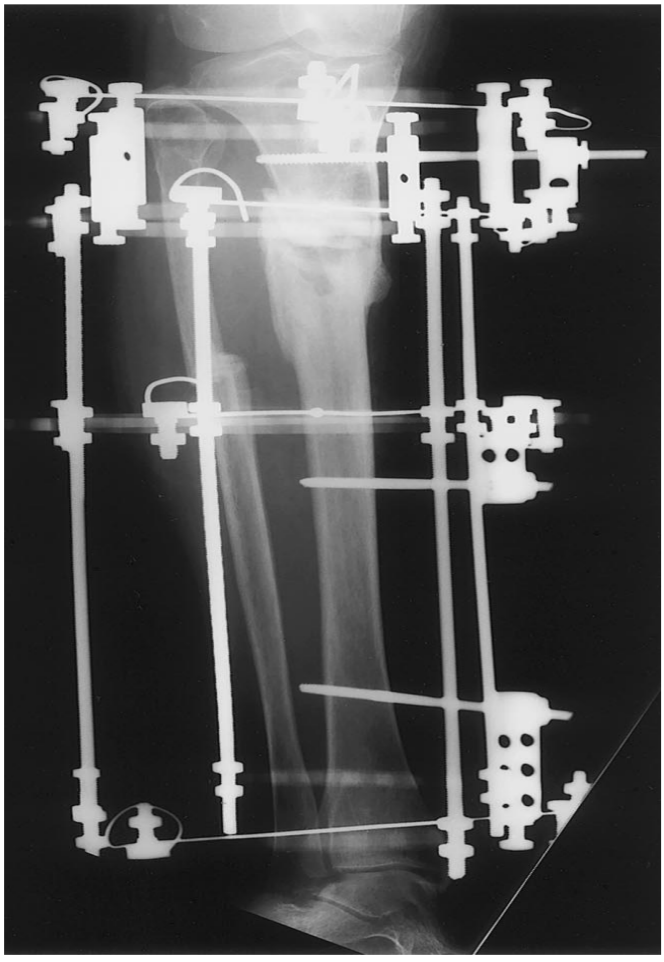


Figure 6: *Ring external fixator (Taljanovic et al., 2003).*

Biomaterials Currently Used in Fracture Fixation

Titanium and stainless steel remain the dominant metals used in fracture fixation due to their corrosion resistance, reduced stress shielding and biocompatibility. Titanium however can be expensive. Stainless steel is cheaper but presents more toxicity concerns. Polymers such as PMMA, PGA, PLLA, and PEEK are also used in fracture fixation as they have Young's moduli close to cancellous bone, reducing stress shielding. Concerns limiting polymer usage include changes in mechanical behaviour, micromotion and tribological issues. Carbon fibre-reinforced PEEK (CFR-PEEK) is currently the most developed composite, offering radiolucency, lower stiffness, and no metal ion release (Zhang et al., 2022). *Table 3* compares materials currently used in fracture fixation.

Table 3: *Comparison of Different Materials Used for Fracture Fixation, based on Kim et al., 2020.*

Material Type	Advantages	Disadvantages	Clinical Applications
Metals	• High mechanical stability • Long-term durability	• Stress shielding • May cause allergic reactions • May require surgical removal	• Used in most fracture fixation applications including plates, screws, IM nails/rods
Polymers	• No stress shielding • Can be bioabsorbable, mitigating surgical removal • Radiolucency allows better post-operative imaging	• Lower mechanical strength than metals • Some polymers degrade unpredictably • Unsuitable for high-load applications	• Bioabsorbable screws, plates, and pins • Used in non-load-bearing bone fixation
Bioceramics	• Osteoconductive • Biodegradable, mitigating surgical removal	• Brittle and prone to fracture under stress • Poor mechanical strength	• Coatings for metallic implants • Used in bone grafting and scaffold materials
Biodegradable metals	• Degrades *in vivo*, mitigating surgical removal • Promotes natural bone regeneration	• Risk of premature degradation or gas formation (magnesium-based) • Limited long-term clinical studies	• Experimental use in bioresorbable screws and plates

A Review of Experimental Testing of Fracture Fixation Plates

Zhang et al. conducted a review of the latest biomechanical research on fracture fixation plates, focusing on experimental testing with simulated or cadaveric bones to assess the strength and stability of various existing fracture fixation plates and screw configurations. *Table 4* presents an overview of their findings. Long-term studies on plate performance however remain scarce, and there is a significant research gap on the tribological behaviour of existing plates. Authors recommend establishing tribological testing standards to reduce risks of debris-induced osteolysis, inflammation, and aseptic loosening (Zhang et al., 2022).

***Table 4**:Overview of experimental testing of fracture fixation plates over the years (Zhang et al., 2022)*

Test Type	Findings	Test Limitations
Compression Testing	Plates with larger stiffness and cross-sectional areas withstand higher loads but increase stress shielding.	Do not account for real-life loading conditions.
Bending Testing	Locking plates show higher bending resistance than non-locking plates. Higher screw density improves bending strength but can lead to bone over-stiffening.	May not reflect actual loading conditions during rehabilitation.
Torsion Testing	Plates with asymmetric hole patterns and variable screw angles show better torsional resistance. Locking plates reduce rotational failure compared to non-locking plates.	Torsional forces *in vivo* are complex and difficult to replicate in experimental setups.
Fatigue Testing	Plates fail at screw-plate interfaces under cyclic loading. Titanium plates show better fatigue resistance than stainless steel. Hybrid plates (metal-polymer) show mixed performance during fatigue testing.	Fatigue properties depend on bone quality, which varies among individuals.

Bone grafts

Bone grafts are implanted to treat fractures, bone defects and non-unions. Autografts remain the gold standard due to their osteogenic, osteoconductive, and osteoinductive properties. As seen in *Table 5*, autografts are the most viable option if the patient has good bone density and thus may be worth it for young and healthy patients despite the invasiveness. Synthetic materials like calcium phosphate-based biomaterials and Bone Morphogenetic Proteins (BMPs) are commonly used, but challenges persist in achieving optimal osteointegration (Wang and Yeung, 2017).

***Table 5:** Comparison of bone graft types (Kim et al., 2020)*

Graft type	Source	Advantages	Disadvantages
Autograft	Patient's Iliac crest	• No immune rejection • High success rate • Osteoinductive and osteoconductive	• Requires good bone density • Harvesting is painful • Harvest site takes long to recover • Limited quantity • Additional surgery required
Allograft	Cadaveric donor bone	• Readily available • No donor site morbidity • No pain in harvest site • Osteoinductive and osteoconductive	• Potential immune response • Risk of disease transmission • Lower osteogenic potential
Xenograft	Non-human species, (e.g. bovine, porcine)	• Abundant supply • Long shelf life	• Possible immune rejection • Requires extensive processing to reduce antigenicity
Synthetic grafts (Allografts)	Man-made materials (e.g. hydroxyapatite, β-TCP, bioactive glass)	• No immune response • Customisable • Biodegradable • Readily available • Easily sterilised	• Lack osteogenic and osteoconductive properties • Slower integration with native bone • Lack required mechanical properties • Require growth factors

Current Treatment Methods

Current treatments for impaired fracture healing include conservative therapies like Low-Intensity Pulsed Ultrasound (LIPUS) and Extracorporeal Shock Wave Therapy (ESWT) (*Figure 7*), as well as surgical procedures including internal and external fixation and bone grafting (*Table 6*). Surgery however remains the most effective, especially for non-unions (Schlickewei et al., 2019).

Table 6 : *Overview of Current Treatment Methods for Fracture Healing based on Schlickewei et al., 2019.*

Type	Method	Description	Application
Conservative Therapy	*Low-Intensity Pulsed Ultrasound (LIPUS)*	Use of mechanical stimulation for facilitate fracture healing; requires daily therapy for 3-6 months.	Hypertrophic non-unions; suitable for patients at high surgical risk.
	Extracorporeal Shock Wave Therapy (ESWT)	Use of high-energy pulses to stimulate osteoblast activity and angiogenesis.	Limited to selective cases of non-infected non-unions.
Surgical Therapy	*Debridement*	Removal of necrotic/infected bone to create a viable environment for bone healing.	Osteomyelitis and cases with non-union requiring removal of devitalised bone.
	Bone Grafting	Surgical bone transplant to provide structural support and biological stimulation to fractured bone.	Fractures with large bone defects or compromised healing environments.
	Internal Fixation	Use of surgical devices to stabilise fractured bone *in vivo.*	Long bone and periarticular fractures.
	External Fixation	Use of rigid frames screwed into fractured bone.	Fracture cases with severe soft tissue damage where internal fixation is not initially possible.
	Two-Stage Therapy	Stage 1: Removal of infection with a temporary antibiotic spacer; Stage 2: Definitive fixation and bone grafting.	Infected or atrophic non-unions where infection control is needed before reconstruction.

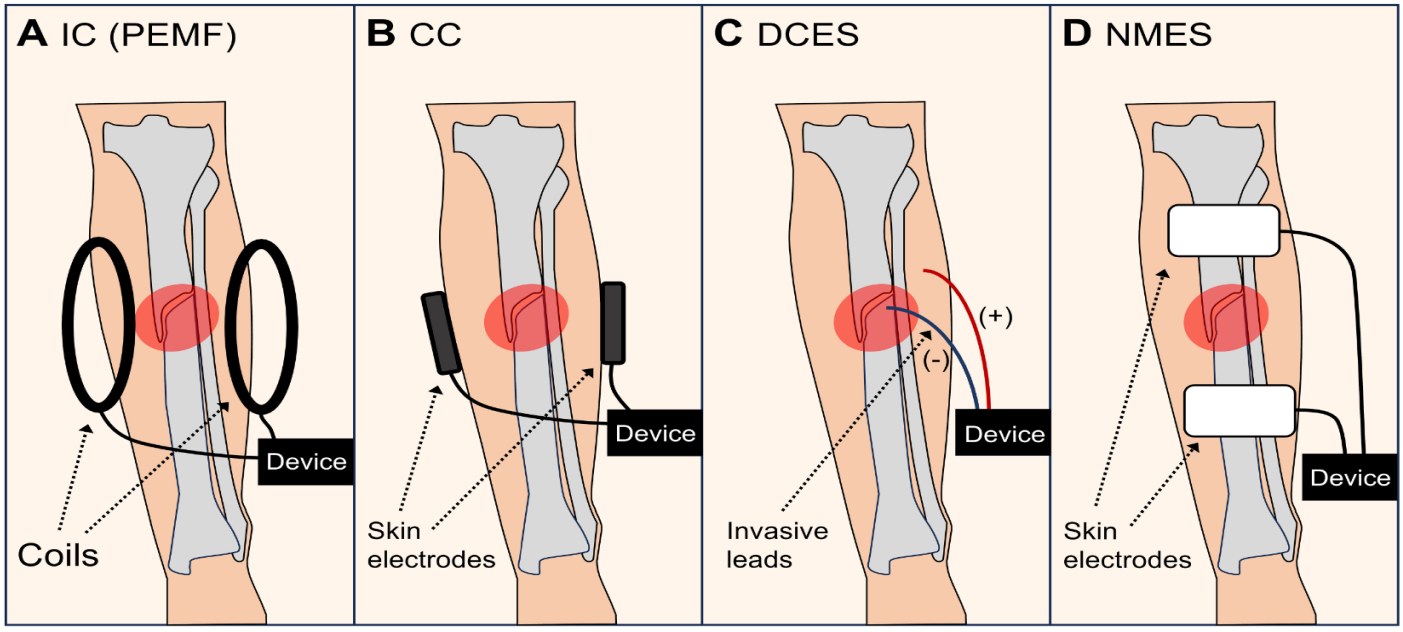


Figure 7: *Electric Shock Wave Therapy - Types of electrical stimulation.* ***(A)*** *inductive coupling,* ***(B)*** *capacitive Coupling* ***(C)*** *Direct electrical stimulation* ***(D)*** *Neuromuscular electrical stimulation (Ganse, 2024).*

Risk Factors for Impaired Fracture Healing

Factors that may cause difficulties in fracture healing and lead to impaired bone regeneration could be physiological, pathological or anatomical (See *Table 7*). About 10-15% of bone fracture cases fail to heal, leading to non-unions (Einhorn, 1998). Non-unions can be hypertrophic (with active bone formation but mechanical instability) or atrophic (without bone healing due to insufficient vascular supply or infection) (See *Figure 8*). Non-unions and delayed unions (slower than expected healing after fixation) lead to revision surgeries and prolonged treatment, presenting a major socio-economic burden as it affects patients' physical and mental health (Schlickewei et al., 2019).

Table 7: *Summary of Risk Factors for Impaired Fracture Healing, Modified According to Andrzejowski and Giannoudis, 2019.*

Physiological Factors	Pathological Factors	Anatomical Factors
Age	Smoking	Open Fracture (Initial Bone Loss and Soft Tissue Injury)
Gender (Male)	Alcohol	Displaced Fracture
High Body Mass	Osteoporosis	Wedge and Multi-Fragmentary Fracture Pattern
Medication (Insulin, Chemotherapeutics, Steroids, etc.)	Diabetes	Injury Location (Tibia)
	Peripheral Vascular Disease	Soft Tissue Damage - Vascularization
	Radiotherapy	Open Reduction - Quality of Primary Osteosynthesis
	Renal Insufficiency	Compartment Syndrome
	Chronic Inflammatory Disease	Presence of Fracture Gap Post-Surgery
		Poor Mechanical Stability by Initial Implant

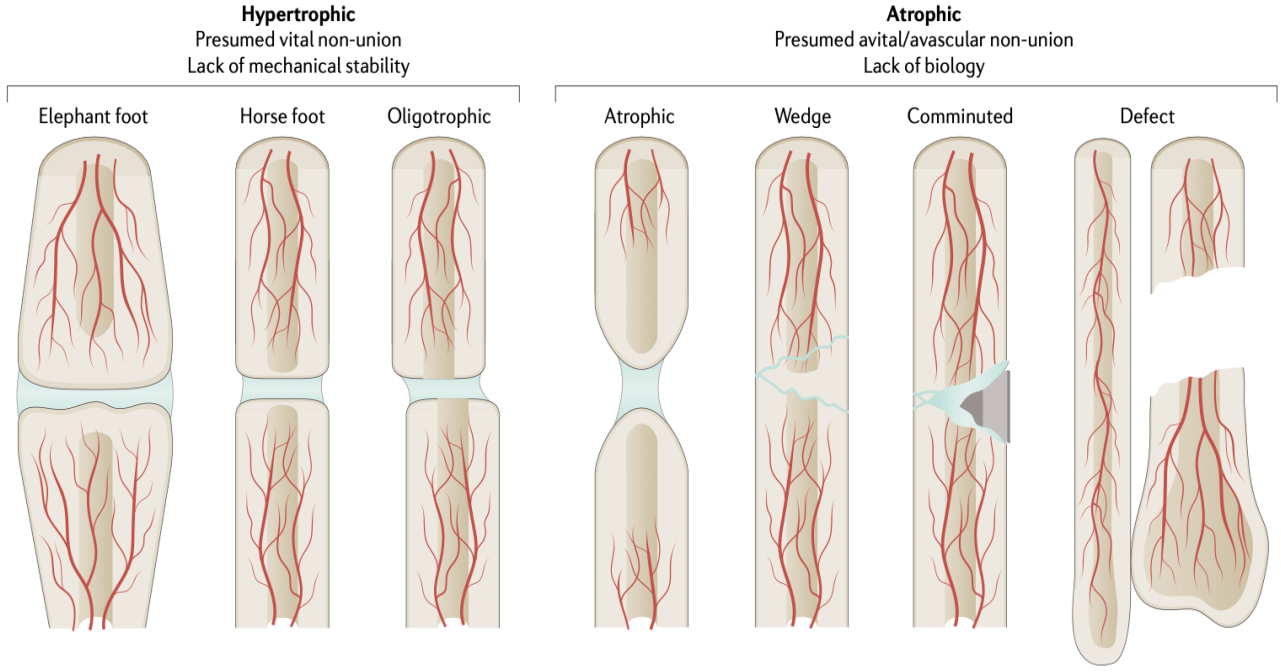


Figure 8: *Hypertrophic and Atrophic Non-Unions (Wildemann et al., 2021).*

Issues Associated with Fracture Healing

It is clear from *Table 8* that tibial and femoral fractures are the most associated with fracture healing complications.

Table 8: *Problems Associated with Fracture Fixation.*

Reference	*Problem*	*Frequency*	*Common Locations*
(Saul et al., 2023)	**Non-Union**	• 2–10% of fractures. • Up to 50% in open tibial fractures.	• Tibia • Femur • Humerus • Clavicles
(Saul et al.,2023)	**Delayed Union**	• Affects around 600,000 fractures in the U.S per year.	• Tibia • Femur
(Saul et al., 2023); (Andrzejowski and Giannoudis, 2019)	**Fracture Related Infections (Osteomyelitis)**	• Up to 30% of open fractures.	• Tibia • Pelvis • Femur • High-energy trauma
(Kim et al., 2020)	**Implant Failure**	• Common in elderly patients or patients with osteoporosis. • Common in load-bearing internal fixation.	• Tibia • Femur • Humerus
(Marsell and Einhorn, 2011)	**Malalignment**	• Common in complex comminuted fractures.	• Ankle • Femur • Radius
(Andrzejowski and Giannoudis, 2019); (Glatt, Evans and Tetsworth, 2017)	**Pin-Track Infections**	• Reported in 5–30% of cases.	• Tibia • Radius • Femur
(Zhu et al., 2023)	**Refracture**	• Reported in 2-5% of fracture cases • Risk increases in osteoporotic patients	• Tibia • Femur

ADVANCEMENTS IN RESEARCH AND TECHNOLOGY

Recent advancements (Summarised in *Table 9*) show promising contributions to fracture healing but currently lack sufficient quality human clinical trials according to Ganse (2024).

Table 9: *Recent advancements in research and technology in fracture healing (Ganse, 2024).*

Category	Type	Advantages	Disadvantages
Mechanical Stimulation	*Axial Micromovement*	• Reduces healing time by 21-23% in tibial fractures. • Improves fracture site stability.	• Requires external fixators (can cause discomfort, infections, and mechanical failures).
	Whole-Body Vibration (WBV)	• Non-invasive.	• Lacks standardised optimal settings.
	Shock Wave Therapy (ESWT)	• Stimulates mechanotransduction and growth factor release. • Beneficial for delayed and non-unions	• Lack of quality human clinical trials.
	Ultrasound (LIPUS)	• Non-invasive. • Some early studies suggested faster healing.	• Recent meta-analyses question effectiveness.
Electromagnetic Stimulation	*Pulsed Electromagnetic Fields (PEMF)*	• Proven to reduce healing time by 17-21%. • Promotes angiogenesis and bone formation.	• Lacks standardised optimal settings.
	Capacitive Coupling (CC)	• Non-invasive. • May improve vertebral fracture healing.	• Limited penetration depth in long bones due to soft tissue interference.
	Direct Current Electrical Stimulation (DCES)	• Can reduce healing time by up to 30%.	• Invasive
	Neuromuscular Electrical Stimulation (NMES)	• Can stimulate muscle contraction, indirectly affecting bone healing. • Useful in immobilised patients.	• Limited human clinical trials. • Difficult to standardise due to muscle variability.
Pharmacologic Interventions	*Growth Factors (BMP-2, FGF-2, PDGF)*	• Enhances callus formation. • May replicate fracture acceleration seen in TBI patients.	• Lack of quality human clinical trials. • Injectable growth factors may have same benefit.
	Parathyroid Hormone (PTH)	• Strongest evidence for healing acceleration. • Reduces healing time by 30% in some fractures.	• Higher doses do not always improve healing. • Effectiveness varies by fracture type.
	Other Medications (Statins, Bisphosphonates, Beta-Blockers)	• Some studies suggest positive effects on bone metabolism. • Can be used in osteoporosis treatment.	• Potential systemic side effects. • Effectiveness varies. • Not widely studied for fracture healing.
Heat-Based Therapies	*Low-Level Laser Therapy (LLLT)*	• Non-invasive.	• Limited human clinical evidence.
	High-Intensity Laser Therapy (HIL)	• Enhances bone regeneration in animal models. • May improve healing rate.	• Limited human clinical evidence.

CONCLUSIONS

Persistent challenges in fracture fixation such as non-unions, stress shielding, and implant-related infections highlight the need to improve current fixation strategies. Emerging technologies including LIPUS and ESWT show promise in complementing current methods but require further validation through performing more quality clinical human trials. Future research should focus on establishing standardised tribological tests—particularly debris generation—to reduce post-operative complications in current fixation interventions and to improve implant longevity. Future research should also include long-term clinical evaluations of emerging interventions and biocompatible materials to ensure better patient outcomes in fracture healing.

REFERENCE LIST

Andrzejowski, P. and Giannoudis, P.V. (2019). The 'diamond concept' for long bone non-union management. *Journal of Orthopaedics and Traumatology*, [online] 20(1). doi:https://doi.org/10.1186/s10195-019-0528-0.

Bigham-Sadegh, A. and Oryan, A. (2014). Basic concepts regarding fracture healing and the current options and future directions in managing bone fractures. *International Wound Journal*, [online] 12(3), pp.238–247. doi:https://doi.org/10.1111/iwj.12231.

Danis, D.R. (1949). *Théorie et pratique de l'ostéosynthèse: par Robert Danis...* Masson (Niort.

Einhorn, T.A. (1998). The Cell and Molecular Biology of Fracture Healing. *Clinical Orthopaedics and Related Research®*, [online] 355. Available at: https://journals.lww.com/clinorthop/fulltext/1998/10001/the_cell_and_molecular_biology_of_fracture_healing.3.aspx.

Field, J.R., Hearn, T.C. and Caldwell, C.B. (1997). Bone Plate Fixation: An Evaluation of Interface Contact Area and Force of the Dynamic Compression Plate (DCP) and the Limited Contact Dynamic Compression Plate (LCDCP) Applied to Cadaveric Bone. *Journal of Orthopaedic Trauma*, [online] 11(5). Available at: https://journals.lww.com/jorthotrauma/fulltext/1997/07000/bone_plate_fixation__an_evaluation_of_interface.13.aspx.

Frost, H.M. (1987). Bone 'mass' and the 'mechanostat': A proposal. *The Anatomical Record*, 219(1), pp.1–9. doi:https://doi.org/10.1002/ar.1092190104.

Ganse, B. (2024). Methods to accelerate fracture healing–a narrative review from a clinical perspective. *Frontiers in Immunology*, 15, p.1384783.

Giannoudis, P.V., Einhorn, T.A. and Marsh, D. (2007). Fracture healing: The diamond concept. *Injury*, 38, pp.S3–S6. doi:https://doi.org/10.1016/s0020-1383(08)70003-2.

Glatt, V., Evans, C.H. and Tetsworth, K. (2017). A concert between biology and biomechanics: The influence of the mechanical environment on bone healing. *Frontiers in Physiology*, [online] 7. doi:https://doi.org/10.3389/fphys.2016.00678.

Hannigan, G.D., Pulos, N., Grice, E.A. and Mehta, S. (2015). Current Concepts and Ongoing Research in the Prevention and Treatment of Open Fracture Infections. *Advances in Wound Care*, 4(1), pp.59–74. doi:https://doi.org/10.1089/wound.2014.0531.

Kim, T., See, C.W., Li, X. and Zhu, D. (2020). Orthopedic implants and devices for bone fractures and defects: Past, present and perspective. *Engineered Regeneration*, [online] 1, pp.6–18. doi:https://doi.org/10.1016/j.engreg.2020.05.003.

Marsell, R. and Einhorn, T.A. (2011). The Biology of Fracture Healing. *Injury*, 42(6), pp.551–555. doi:https://doi.org/10.1016/j.injury.2011.03.031.

Oryan, A., Alidadi, S., Moshiri, A. and Bigham-Sadegh, A. (2014). Bone morphogenetic proteins: A powerful osteoinductive compound with non-negligible side effects and limitations. *BioFactors*, 40(5), pp.459–481. doi:https://doi.org/10.1002/biof.1177.

Oryan, A., Kamali, A., Moshiri, A. and Baghaban Eslaminejad, M. (2017). Role of Mesenchymal Stem Cells in Bone Regenerative Medicine: What Is the Evidence? *Cells Tissues Organs*, 204(2), pp.59–83. doi:https://doi.org/10.1159/000469704.

Parra-Cabrera, G., Pérez-Cano, F.D. and Jiménez-Delgado, J.J. (2022). Fracture pattern projection on 3D bone models as support for bone fracture simulations. *Computer Methods and Programs in Biomedicine*, 224, pp.106980–106980. doi:https://doi.org/10.1016/j.cmpb.2022.106980.

Saul, D., Menger, M.M., Ehnert, S., Nüssler, A.K., Histing, T. and Laschke, M.W. (2023). Bone Healing Gone Wrong: Pathological Fracture Healing and Non-Unions—Overview of Basic and Clinical Aspects and Systematic Review of Risk Factors. *Bioengineering*, 10(1), p.85. doi:https://doi.org/10.3390/bioengineering10010085.

Schlickewei, C.W., Kleinertz, H., Thiesen, D.M., Mader, K., Priemel, M., Frosch, K.-H. and Keller, J. (2019). Current and Future Concepts for the Treatment of Impaired Fracture Healing. *International Journal of Molecular Sciences*, [online] 20(22). doi:https://doi.org/10.3390/ijms20225805.

Taljanovic, M.S., Jones, M.D., Ruth, J.T., Benjamin, J.B., Sheppard, J.E. and Hunter, T.B. (2003). Fracture Fixation. *RadioGraphics*, 23(6), pp.1569–1590. doi:https://doi.org/10.1148/rg.236035159.

Uhthoff, H.K., Poitras, P. and Backman, D.S. (2006). Internal plate fixation of fractures: short history and recent developments. *Journal of Orthopaedic Science*, [online] 11(2), pp.118–126. doi:https://doi.org/10.1007/s00776-005-0984-7.

Wang, W. and Yeung, K.W.K. (2017). Bone grafts and biomaterials substitutes for bone defect repair: A review. *Bioactive Materials*, 2(4), pp.224–247. doi:https://doi.org/10.1016/j.bioactmat.2017.05.007.

Wildemann, B., Ignatius, A., Leung, F., Taitsman, L.A., Smith, R.M., Pesántez, R., Stoddart, M.J., Richards, R.G. and Jupiter, J.B. (2021). Non-union bone fractures. *Nature Reviews Disease Primers*, 7(1). doi:https://doi.org/10.1038/s41572-021-00289-8.

Zhang, S., Patel, D., Brady, M., Gambill, S., Theivendran, K., Deshmukh, S., Swadener, J., Junaid, S. and Leslie, L.J. (2022). Experimental testing of fracture fixation plates: A review. *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, 236(9), pp.1253–1272. doi:https://doi.org/10.1177/09544119221108540.

Zhu, Y., Hu, J., Zhan, T., Zhu, K. and Zhang, C. (2023). Refracture after plate removal of midshaft clavicle fractures after bone union—incidence, risk factors, management and outcomes. *BMC Musculoskeletal Disorders*, 24(1). doi:https://doi.org/10.1186/s12891-023-06391-0.

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-In an Executive Summary or an Abstract it is not helpful to your readers to use abbreviations without explaining (stating) them.

-Why do you think stainless steel and titanium are toxic?

NOTE:

Please note that in the body of the text and in tables the initials or first names of authors are not used.

When putting images in reports please make sure you keep the height to width ratio constant and therefore do not distort the image.

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